

Science leaderboard

Last month we looked at human discoveries in the realm of science. This month it's the natural world.

**More Bits per photon**

Researchers have found a way to encode 2.05 Bits per photon using twisted light. More here: <http://dg.it/MoreBits>

Going nuts for space

Over the years humanity has come up with some very crazy ways to achieve space travel. Crazier still – some of them might work.

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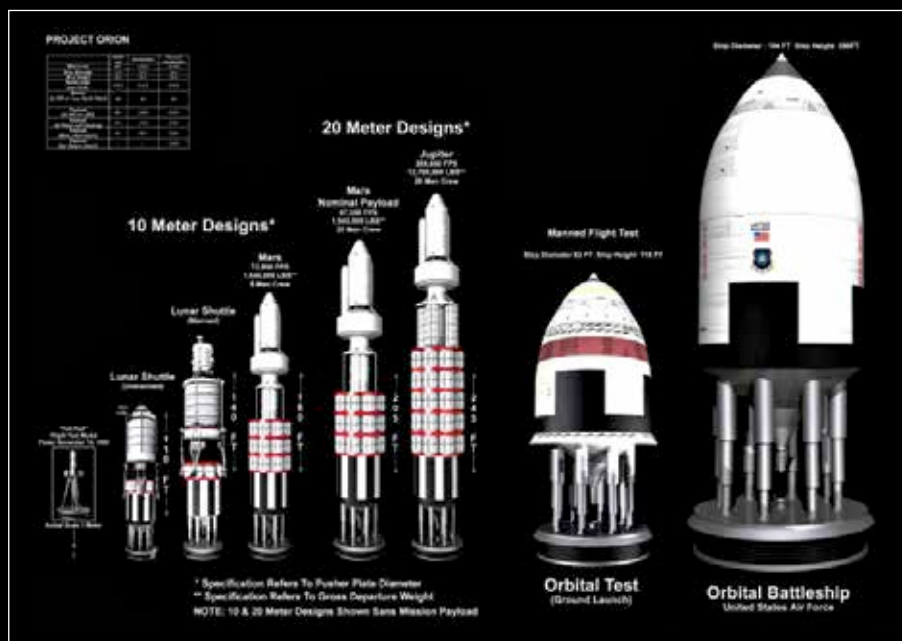
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The line between insanity and genius is very thin. And some scientists want to cross it using space catapults, nuclear bombs and laser beams. The strange thing however is that while their ideas may seem outlandish and insane, they are all scientifically sound. Historically this combination of traits has led to some of the greatest discoveries, and the most horrific events (Nazi eugenics). So we can only be glad that most of these ideas have fallen by the wayside and reduced to comical anecdotes.

When it comes to space travel however the crazier ideas seem to be the best ones. After all, who would have thought that sitting on a needle tip of a 110 meters long intercontinental ballistic missile would take us to the moon? That is what the Saturn V launch vehicle was - a 3000 ton bullet fuelled by 1.2 million litres of high octane fuel that exploded on the face of the Earth in order to successfully carry humans beyond low Earth orbit. Keeping with that tradition, we bring together some of the craziest ideas ever proposed for human space flight.

Project Orion a.k.a Nuclear Farts

On the face of it Project Orion was the most legitimate project to further space travel since the Apollo space program. The project was officially kicked off in 1958 with great minds such as Freeman Dyson leading its charge and was under



Successive nuclear blasts propel the craft into space at high efficiencies.

serious consideration by the National Aeronautical Space Agency for nearly a decade. The seeds of its scientific premise were rooted in classical physics and had been established in the late 19th century by Russian explosive genius Nikolai Kibalchich and German engineer Hermann Ganswindt. The idea was simple - high impact explosions that occur outside the launch vehicle which propel it into space. This form of propulsion allowed for higher fuel efficiency, higher thrusts and higher specific impulse than any design ever conceived. There was only one hitch - it was nuclear.

Project Orion called for the ignition of nuclear fuel as its main propellant. This meant that in order to launch and sustain flight the launch vehicle would basically be exploding nuclear bombs

in the atmosphere on its way to breaking orbit. The engineering practicalities were well understood and well within the technology of the day. The possibilities of this launch vehicle were so phenomenal that it was a serious contender for interplanetary space travel as it could achieve the greatest speeds with the least fuel payloads. Almost all aspects of the project were tested and proven, apart from the nuclear propulsion tests. Today thanks to computer simulations we know the design would have worked but back then there was simply no place to conduct hundreds of nuclear blasts to see if the math and physics would hold up. Thankfully, with the 1963 Partial Test Ban Treaty it was no longer legally possible to use nuclear technology in orbit and Project Orion was shelved.